PCA and Dimensionality Reduction

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises**

A health study surveys 5,000 adults and finds that people who sleep 8 hours per night have significantly lower blood pressure than those who sleep 5 hours. (a) Can researchers conclude that increasing sleep causes a reduction in blood pressure? (b) Delineate the scope of inference using the 2×2 selection/assignment matrix.

(a) Causal Inference: No, cause-and-effect cannot be established because this is an observational study where sleep duration was not randomly assigned. Plausible confounding variables (e.g. job stress, diet, exercise) are associated with sleep duration and independently affect blood pressure. (b) Scope of Inference:Were individuals randomly selected from the adult population? Yes $\Rightarrow$ Inferences can be generalized to the broader adult population.Were treatments randomly assigned? No $\Rightarrow$ Inference.

2 **[Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises**

Level 3 · Advanced Find the full interval of convergence (including endpoint behavior) for the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n^2 \cdot 4^n}$.

Apply the Ratio Test: $L = \lim_{n\to\infty} \left| \frac{(2x-1)^{n+1}}{(n+1)^2 4^{n+1}} \cdot \frac{n^2 4^n}{(2x-1)^n} \right| = \lim_{n\to\infty} \left(\frac{n}{n+1}\right)^2 \cdot \frac{|2x-1|}{4} = \frac{|2x-1|}{4}$ Require x: $2x-1=4$, giving $\sum_{n=1}^{\infty} \frac{4^n}{n^2 4^n} = \sum_{n=1}^{\infty} \frac{1}{n^2}$, which converges by the p-series test ($p=2>1$). Endpoint $x=-3/2$: $2x-1=-4$, giving.

3 **[Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises**

Level 2 · MediumDetermine whether the polynomials $p_1(x) = 1 + x$, $p_2(x) = x + x^2$, and $p_3(x) = 1 + x^2$ form a basis for $\mathcal{P}_2$ (the vector space of polynomials of degree at most 2).

Step 1 · Coordinate vectors: The standard basis for $\mathcal{P}_2$ is $\mathcal{E} = \{1, x, x^2\}$, so $\dim(\mathcal{P}_2) = 3$. The coordinate vectors are: $[p_1]_{\mathcal{E}} = (1;1;0), \quad [p_2]_{\mathcal{E}} = (0;1;1), \quad [p_3]_{\mathcal{E}} = (1;0;1).$ Step 2 · Test independence via determinant: $\det(1,0,1;1,1,0;0,1,1) = 1(1-0) - 0 + 1(1-0) = 1+1 = 2eq0.$ Since the determinant is non-zero, the three coordinate vectors are linearly.

4 **[Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises**

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix $A = (4,1;2,3)$. Find the matrix representation $[T]_{\mathcal{B}}$ relative to the basis $\mathcal{B} = \{(1;1), (-1;2)\}$ via similarity transformation $P^{-1}AP$.

Step 1: Construct change-of-basis transition matrix P: $P = (1, -1; 1, 2), \quad \det(P) = 2 - (-1) = 3 \Rightarrow P^{-1} = \frac{1}{3}(2, 1; -1, 1).$ Step 2: Compute AP: $AP = (4,1;2,3)(1, -1; 1, 2) = (4(1)+1(1), 4(-1)+1(2); 2(1)+3(1), 2(-1)+3(2)) = (5, -2; 5, 4).$ Step 3: Compute $[T]_{\mathcal{B}} = P^{-1}(AP)$.

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5 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

An agronomist wants to test whether a new organic fertilizer increases tomato yield compared to a standard commercial fertilizer. Explain how the agronomist must incorporate the four principles of experimental design: Comparison, Random Assignment, Control, and Replication.

1. Comparison: Use two treatment groups (organic fertilizer vs. standard fertilizer control) grown concurrently under the same environmental conditions. 2. Random Assignment: Assign tomato plants to fertilizer treatments using a random number generator to balance unmeasured variables (e.g. baseline soil nutrient variations, microclimates). 3. Control: Keep all extraneous variables identical for both groups: sunlight exposure, watering schedule, soil volume, and pest control. 4. Replication: Apply each fertilizer to.

6 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Let $\mathbf{u} = (1, -2, 2, 0)^T$ and $\mathbf{v} = (2, 1, 0, -2)^T$ in $\mathbb{R}^4$. Compute $||\mathbf{u}||$, $||\mathbf{v}||$, the inner product $\mathbf{u} \cdot \mathbf{v}$, the cosine of the angle θ between them, and verify the Cauchy–Schwarz inequality.

1. Norms: $||\mathbf{u}|| = \sqrt{1^2 + (-2)^2 + 2^2 + 0^2} = \sqrt{1 + 4 + 4 + 0} = \sqrt{9} = 3.$ $||\mathbf{v}|| = \sqrt{2^2 + 1^2 + 0^2 + (-2)^2} = \sqrt{4 + 1 + 0 + 4} = \sqrt{9} = 3.$ 2. Inner Product: $\mathbf{u} \cdot \mathbf{v} = (1)(2) + (-2)(1) + (2)(0) + (0)(-2) = 2 - 2 + 0 + 0 = 0.$ 3. Angle: $\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||||\mathbf{v}||} = \frac{0}{3 \cdot 3} = 0 \Rightarrow \theta = \frac{\pi}{2}$ rad (90°). 4. Cauchy–Schwarz Verification.

7 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 3 · Advanced A particle moves along a curve defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the value of $\frac{d^2y}{dx^2}$ at $t = 2$ and determine whether the curve is concave up or concave down at this point.

First find the components: $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 3$. The first derivative is $\frac{dy}{dx} = \frac{3t^2 - 3}{2t} = \frac{3}{2}t - \frac{3}{2t}$. Differentiating with respect to t: $\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2} + \frac{3}{2t^2}$. Then $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(dy/dx)}{dx/dt} = \frac{\frac{3}{2} + \frac{3}{2t^2}}{2t}$. At $t = 2$: numerator is $\frac{3}{2} + \frac{3}{8} = \frac{15}{8}$; denominator is $2(2) = 4$. Thus.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

A school district with 12 high schools wants to sample 300 teachers. Compare: (a) Stratified sampling (taking an SRS of 25 teachers from each of the 12 schools) vs. (b) Cluster sampling (randomly selecting 2 schools and surveying all 150 teachers in each). Discuss sampling variability and logistical practicality.

(a) Stratified Sampling: Guarantees representation from every high school. Because schools may have differing cultures and student demographics (homogeneous within schools, heterogeneous between schools), stratifying reduces sampling variability between strata, providing a more precise district-wide estimate. (b) Cluster Sampling: Highly practical and cost-effective because surveyors only visit 2 physical campuses. However, if individual schools differ substantially from one another, cluster sampling introduces muc.

9 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Evaluate the integral $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

Step 1: Sketch and identify the region D: The given bounds are $y \le x \le 1$ for $0 \le y \le 1$. This is the triangle bounded by $x = y$, $x = 1$, and $y = 0$. Step 2: Reverse order to integrate with respect to y first: In the reversed order, $0 \le y \le x$ for $0 \le x \le 1$: $\int_0^1 \int_0^x e^{x^2}\,dy\,dx$. Step 3: Evaluate inner integral: $\int_0^x e^{x^2}\,dy = [ye^{x^2}]_{y=0}^{y=x} = xe^{x^2}$. Step 4: Evaluate outer integral using substitution $u = x^2$, $du = 2x\,dx$.

10 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

Compute second derivative: $f(x) = x^{-1} \Rightarrow f'(x) = -x^{-2} \Rightarrow f''(x) = 2x^{-3} = \frac{2}{x^3}$. On $[1, 3]$, $|f''(x)| = \frac{2}{x^3}$ is decreasing, so its maximum occurs at $x = 1$: $K = \frac{2}{1^3} = 2$. Using the Trapezoidal error bound with $a = 1, b = 3$: $|E_T| \le \frac{K(b-a)^3}{12n^2} = \frac{2(3-1)^3}{12n^2} = \frac{2(8)}{12n^2} = \frac{16}{12n^2} = \frac{4}{3n^2}$. Require.

11 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.

Compute derivatives: $\frac{dx}{dt} = 3\cos^2 t(-\sin t) = -3\cos^2 t \sin t$ $\frac{dy}{dt} = 3\sin^2 t(\cos t) = 3\sin^2 t \cos t$ Form the integrand: $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t = 9\cos^2 t \sin^2 t(\cos^2 t + \sin^2 t) = 9\cos^2 t \sin^2 t$ Since $t \in [0, \pi/2]$, $\cos t \ge 0$ and $\sin t \ge 0$, so the square root is: $\sqrt{9\cos^2 t \sin^2 t} = 3\cos t \sin t = \frac{3}{2}\sin(2t)$ Integrate on $[0, \pi/2]$.

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12 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 3 · AdvancedA university alumni association randomly selects 500 graduates from its directory and sends them a survey asking for their current annual salary. Only 95 completed surveys are returned, yielding an average reported salary of 115,000. (a) Name the primary source of bias present. (b) Explain the likely direction of the bias on the estimated mean salary. (c) Explain whether the results can be generalized to all graduates of the university.

(a) Source of Bias: Nonresponse bias. A representative random sample was selected, but 81% (405 of 500) of selected individuals failed to respond. (b) Direction of Bias: The estimate is likely an overestimate (upward bias). Graduates with high incomes and prestigious careers are substantially more eager and proud to report their earnings, while graduates who are unemployed or earn lower wages are far less likely to respond. (c) Generalizability: No, the results cannot be generalized to all graduates because the mas.

13 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of c that satisfy the theorem.

Step 1 · Check Hypotheses: $f(x)$ is continuous on $[1, 4]$ and differentiable on $(1, 4)$ since the only discontinuity $x = 0$ lies outside $[1, 4]$. The hypotheses of MVT hold. Step 2 · Compute Average Rate of Change: $\frac{f(4) - f(1)}{4 - 1} = \frac{(4 + 1/4) - (1 + 1)}{3} = \frac{17/4 - 2}{3} = \frac{9/4}{3} = \frac{3}{4}$. Step 3 · Solve $f'(c) = \frac{3}{4}$.

14 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Find the least-squares solution $\hat{\mathbf{x}}$ to the inconsistent system $A\mathbf{x} = \mathbf{b}$, where

$A = (1, 1; 1, 2; 1, 3), \qquad \mathbf{b} = (1; 2; 4).$

Step 1: Set up normal equations $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$: $A^T A = (1, 1, 1; 1, 2, 3)(1, 1; 1, 2; 1, 3) = (3, 6; 6, 14)$. $A^T\mathbf{b} = (1, 1, 1; 1, 2, 3)(1; 2; 4) = (7; 17)$. Step 2: Invert $A^T A$ and solve: $\det(A^T A) = 3(14) - 6(6) = 42 - 36 = 6$. $(A^T A)^{-1} = \frac{1}{6}(14, -6; -6, 3)$. .

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

Compute the divergence: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$ Apply the Divergence Theorem: $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$ Convert to spherical coordinates: $x^2 + y^2 + z^2 = \rho^2$, $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$.

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16 **[Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

The commute times for employees at a large tech company have a strongly right-skewed distribution with mean $\mu = 38$ minutes and standard deviation $\sigma = 15$ minutes. A random sample of 49 employees is selected. (a) Describe the shape, mean, and standard error of the sampling distribution of the sample mean $\bar{x}$. (b) Calculate $P(\bar{x} > 42 \text{ minutes})$.

(a) Sampling Distribution Characteristics:Shape: By the Central Limit Theorem, since $n = 49 \geq 30$, the sampling distribution of $\bar{x}$ is approximately normal, despite the right skewness of the population.Mean: $\mu_{\bar{x}} = \mu = \mathbf{38}$ minutes.Standard Error: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{49}} = \frac{15}{7} \approx \mathbf{2.143}$ minutes.(b) Probability Calculation: Standardize: $z = \frac{42 - 38}{2.143} = \frac{4}{2.143} \approx 1.87$. .

17 **[Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization**

Level 2 · MediumEvaluate the trigonometric limit: $\lim_{x \to 0} \frac{1 - \cos(3x)}{x^2}$.

Multiply numerator and denominator by the conjugate $1 + \cos(3x)$:

$\lim_{x \to 0} \frac{(1 - \cos(3x))(1 + \cos(3x))}{x^2(1 + \cos(3x))} = \lim_{x \to 0} \frac{1 - \cos^2(3x)}{x^2(1 + \cos(3x))} = \lim_{x \to 0} \frac{\sin^2(3x)}{x^2(1 + \cos(3x))}$. Re-group terms to leverage $\lim_{u \to 0} \frac{\sin u}{u} = 1$: $= \lim_{x \to 0} \left[\left(\frac{\sin(3x)}{3x} \right)^2 \cdot 3^2 \cdot \frac{1}{1 + \cos(3x)} \right]$. Evaluating individual limits.

18 **[Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

Here $f(x) = 2x$ and $f'(x) = 2$. The arc length element is $ds = \sqrt{1 + (f'(x))^2}\, dx = \sqrt{1 + 2^2}\, dx = \sqrt{5}\, dx$. The surface area formula gives: $S = 2\pi \int_0^3 f(x)\sqrt{1 + [f'(x)]^2}\, dx = 2\pi \int_0^3 (2x)\sqrt{5}\, dx = 4\sqrt{5}\pi \left[\frac{x^2}{2} \right]_0^3 = 4\sqrt{5}\pi \left(\frac{9}{2} \right) = 18\sqrt{5}\pi$ Verification via cone geometry: The shape is a right circular cone with radius $r = f(3) = 6$ and slant height $l = \sqrt{3^2 + 6^2} = \sqrt{45} = 3\sqrt{5}$. Lateral are.

19 **[Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Take the natural logarithm of both sides: $\ln(3^{2x+1}) = \ln(5^{x-2})$. Apply the power law of logarithms: $(2x + 1)\ln 3 = (x - 2)\ln 5$. Distribute: $2x\ln 3 + \ln 3 = x\ln 5 - 2\ln 5$. Collect all terms containing x on one side: $2x\ln 3 - x\ln 5 = -\ln 3 - 2\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = -(\ln 3 + 2\ln 5)$. Simplify logarithmic coefficients: $2\ln 3 = \ln 9$ and $2\ln 5 = \ln 25$: $x(\ln 9 - \ln 5) = -(\ln 3 + \ln 25) \Rightarrow x\ln\left(\frac{9}{5}\right) = -\ln(75)$. Solve for x.

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20 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Use Green's Theorem to evaluate $\oint_C (y^2 - 7y)\,dx + (2xy + 2x)\,dy$, where C is the positively oriented boundary of the region D enclosed between the parabolas $y = x^2$ and $x = y^2$.

1. Compute the 2D curl: With $P = y^2 - 7y$ and $Q = 2xy + 2x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = (2y + 2) - (2y - 7) = 2 - (-7) = 9$. 2. Intersections of boundary curves: $y = x^2$ and $x = y^2 \Rightarrow x = x^4 \Rightarrow x = 0, 1$. Region D is described by $0 \le x \le 1$ and $x^2 \le y \le \sqrt{x}$. 3. Apply Green's Theorem.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A pharmaceutical company is testing a new drug to reduce blood pressure. They conduct a clinical trial with $n = 100$ participants. The sample mean reduction in systolic blood pressure is $\bar{x} = 15$ mmHg with a sample standard deviation of $s = 8$ mmHg. The company plans to construct a 95% confidence interval for the true mean reduction in systolic blood pressure. If the margin of error for this confidence interval is $M = 2$ mmHg, what would be the required sample size, n', to achieve a margin of error of $M' = 1$ mmHg with the same confidence level and an estimated population standard deviation equal to the sample standard deviation?

a. 200

b. 300

c. 400

d. **1600** ✓

The formula for the margin of error (ME) of a confidence interval for a population mean is given by $\text{ME} = z^* \frac{\sigma}{\sqrt{n}}$, where z^* is the critical value for the desired confidence level, σ is the population standard deviation, and n is the sample size. In the initial scenario, we have $n = 100$, $s \approx \sigma = 8$ mmHg, and $M = 2$ mmHg. For a 95% confidence interval, the critical value z^* is approximately 1.96. Using the margin of error formula.

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22 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

A pharmaceutical company is conducting a clinical trial to test the efficacy of a new drug designed to lower blood pressure. They recruit 200 participants and randomly assign them to either the treatment group (receiving the new drug) or the control group (receiving a placebo). After 12 weeks, they measure the reduction in systolic blood pressure for each participant. The company wants to perform a hypothesis test to determine if there is a significant difference in blood pressure reduction between the two groups. The trial is designed with 20 participants in the treatment group and 180 participants in the control group. If the standard deviation of blood pressure reduction in the treatment group is $\sigma_1 = 15$ mmHg and in the control group is $\sigma_2 = 12$ mmHg, and they are planning to use a two-sample t-test, what is the calculated pooled standard deviation s_p if equal variances are assumed, given the following summary statistics for blood pressure reduction (in mmHg): $\bar{x}_1 = 10$ mmHg for the treatment group and $\bar{x}_2 = 7$ mmHg for the control group?

a. $\sqrt{\frac{15^2 + 12^2}{2}}$

b. $\sqrt{\frac{20(15^2) + 180(12^2)}{200}}$ ✓

c. $\frac{20 \times 15 + 180 \times 12}{200}$

d. $\sqrt{\frac{19(15^2) + 179(12^2)}{198}}$

The problem asks for the pooled standard deviation s_p when assuming equal variances for a two-sample t-test. The formula for the pooled variance s_p^2 is given by: $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$. In this problem, we have: - Sample size for treatment group, $n_1 = 20$ - Sample standard deviation for treatment group, $s_1 = 15$ mmHg - Sample size for control group, $n_2 = 180$ - Sample standard deviation for control group, $s_2 = 12$ mmHg - Total degrees of.

23 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$. Then $\int_0^{\pi/2} \sin^6 x\,dx =$

a. $5\pi/16$

b. $3\pi/16$ ✓

c. $5\pi/32$

d. $15\pi/48$

Wallis: $\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{5\pi\cdot 3}{6\cdot 4\cdot 2\cdot 2} = \frac{15\pi}{96} = \frac{5\pi}{32}$. Actually Wallis: $\int_0^{\pi/2} \sin^{2n} x\,dx = \frac{(2n-1)!!}{(2n)!!}\frac{\pi}{2}$. For $n = 3$: $\frac{5\cdot 3\cdot 1}{6\cdot 4\cdot 2}\frac{\pi}{2} = \frac{15}{48}\frac{\pi}{2} = \frac{15\pi}{96} = \frac{5\pi}{32}$.

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24 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Using the reduction formula, evaluate $\int_0^{\frac{\pi}{2}} \cos^6(x)\,dx$.

a. $\frac{5\pi}{16}$

b. $\frac{5\pi}{32}$ ✓

c. $\frac{3\pi}{16}$

d. $\frac{1}{6}$

On $[0, \pi/2]$, $I_n = \frac{n-1}{n} I_{n-2}$ with $I_0 = \pi/2$ and $I_1 = 1$. Applying it repeatedly gives $\frac{5\pi}{32}$.

25 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction formula: $I_n = \int \sin^n x\,dx$ satisfies:

a. $I_n = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$ ✓

b. $I_n = \frac{\sin^{n+1} x}{n+1} + C$

c. $I_n = -n\cos x \sin^{n-1} x + C$

d. $I_n = I_{n-1} \cdot I_1$

Standard reduction: $I_n = -\frac{\cos x \sin^{n-1} x}{n} + \frac{n-1}{n} I_{n-2}$.

26 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction formula for $\int \sin^n x\,dx$ comes from IBP with $u =$

a. $\sin^n x$

b. $\cos x$

c. $\sin^{n-1} x$ ✓

d. x

Write $\sin^n x = \sin^{n-1} x \cdot \sin x$, then IBP.

27 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

Reduction of order: given $y_1 = e^x$ solves $y'' - y = 0$. Second independent solution via $y_2 = v(x)e^x$:

a. $y_2 = xe^x$

b. $y_2 = e^x \ln x$

c. $y_2 = xe^{-x}$

d. $y_2 = e^{-x}$ ✓

$v' + v = 0 \Rightarrow v = Ce^{-2x} \ldots y_2 = e^{-x}$.

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28 [Foundational] • Target: PCA & Dimensionality Reduction Practice Exercises

$\int \sin^4 x\,dx$ using reduction formula:

a. $\frac{3x}{4} + \frac{\sin 2x}{4} + C$

b. $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$ ✓

c. $\frac{x}{2} - \frac{\sin 4x}{8} + C$

d. $\frac{\sin^3 x \cos x}{4} + \frac{\sin x \cos x}{4} + \frac{x}{8} + C$

Double application of half-angle.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the function $f(x) = x^2 - 5x + 6$. What is the derivative of $f(x)$ with respect to x?

a. $2x - 5$ ✓

b. $x - 5$

c. $x^2 - 5$

d. $2x$

To find the derivative of the function $f(x) = x^2 - 5x + 6$, we apply the power rule for differentiation, which states that the derivative of ax^n is anx^{n-1}. The derivative of a constant is 0. Applying the power rule to each term: 1. The derivative of x^2 is $2x^{2-1} = 2x$. 2. The derivative of $-5x$ is $-5 \cdot 1 \cdot x^{1-1} = -5x^0 = -5$. 3. The derivative of the constant 6 is 0. Combining these, the derivative of $f(x)$ is $f'(x) = 2x - 5 + 0 = 2x - 5$.

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30 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?

a. (2, 1)

b. **(2, -1) ✓**

c. (1, 0)

d. (3, 0)

The function is a parabola opening upwards, so its local minimum is also its vertex. The x-coordinate of the vertex of a parabola given by $f(x) = ax^2 + bx + c$ is $x = -b/(2a)$. For $f(x) = x^2 - 4x + 3$, we have $a = 1$ and $b = -4$. Therefore, the x-coordinate of the vertex is $x = -(-4)/(2 * 1) = 4/2 = 2$. To find the y-coordinate, we substitute $x = 2$ into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Thus, the coordinates of the local minimum are (2, -1).

31 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

A parallelogram in the xy-plane is defined by the vectors $\mathbf{u} = (3; 1)$ and $\mathbf{v} = (1; 4)$. What is the area of this parallelogram?

a. 10

b. **11 ✓**

c. 13

d. 14

The area of a parallelogram defined by two vectors $\mathbf{u} = (u_1; u_2)$ and $\mathbf{v} = (v_1; v_2)$ in $\mathbb{R}^2$ is given by the absolute value of the determinant of the matrix formed by these vectors as columns (or rows). In this case, the matrix is $A = (3, 1; 1, 4)$. The determinant of A is $\det(A) = (3)(4) - (1)(1) = 12 - 1 = 11$. Therefore, the area of the parallelogram is $|11| = 11$.

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32 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the vector $\mathbf{v} = (3; 4)$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

a. The vector's direction in the xy-plane.

b. The angle the vector makes with the positive x-axis.

c. **The distance of the point (3,4) from the origin.** ✓

d. The sum of the components of the vector.

The length of a vector $\mathbf{v} = (x; y)$ is given by its magnitude, $= x^2 + y^2$. Geometrically, this corresponds to the distance from the origin $(0, 0)$ to the point (x, y) on the Cartesian plane, as described by the Pythagorean theorem. For $\mathbf{v} = (3; 4)$, the length is $\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. This value represents the distance of the point $(3, 4)$ from the origin.

33 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?

a. **1** ✓

b. 0

c. 2

d. Does not exist

The limit of a function as x approaches a certain value is the value that $f(x)$ gets arbitrarily close to as x gets arbitrarily close to that value. Observing the graph, as x approaches 1 from both the left and the right, the value of $f(x)$ approaches 1. Therefore, $\lim_{x \to 1} f(x) = 1$.

34 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Consider the vector $\mathbf{v} = (-3; 4)$ in $\mathbb{R}^2$. What is the magnitude (or norm) of this vector?

a. **5** ✓

b. 7

c. 1

d. 25

The magnitude (or norm) of a vector $\mathbf{v} = (x; y)$ in $\mathbb{R}^2$ is given by the formula $= x^2 + y^2$. For the given vector $\mathbf{v} = (-3; 4)$, we have $x = -3$ and $y = 4$. Therefore, the magnitude is $= (-3)^2 + (4)^2 = 9 + 16 = 25 = 5$.

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35 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Water pours into an inverted cone (radius 4 m, height 6 m) at 4 m³/min. How fast is the depth rising when it is 4 m? (m/min)

a. $\frac{3}{8\pi}$

b. $\frac{9}{4\pi}$

c. $\frac{9}{16\pi}$ ✓

d. $\frac{1}{4\pi}$

By similar triangles $r = \frac{2h}{3}$, so $V = \frac{4\pi h^3}{27}$. Then $\frac{dV}{dt} = \pi\frac{16}{36}h^2\frac{dh}{dt}$, which gives $\frac{9}{16\pi}$ m/min at $h = 4$.

36 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

Water drains from a cone-shaped tank (radius 3, height 6) at $\frac{dV}{dt} = 2\ \text{m}^3/\text{min}$. Using $V = \frac{1}{3}\pi r^2 h$ with $r = \frac{3}{6}h$, find $\frac{dh}{dt}$ when $h = 2$.

a. $\frac{2}{\pi}$ ✓

b. $\frac{1}{\pi}$

c. $-\frac{2}{\pi}$

d. $\frac{4}{\pi}$

$V = \frac{\pi}{3}(\frac{3}{6})^2h^3 \Rightarrow \frac{dV}{dt} = \pi(\frac{3}{6})^2h^2\frac{dh}{dt}$. Solving at $h = 2$: $\frac{dh}{dt} = \frac{2}{\pi}$ m/min.

37 [Intermediate] • Target: PCA & Dimensionality Reduction Practice Exercises

The slope of $y = 3x - 5$ at any point is:

a. -5

b. 0

c. **3** ✓

d. undefined

Applying Derivatives principles, Slope of a line $y = mx + b$ is $m = 3$. Therefore, the verified correct option is 3.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Circle centred at origin, radius 5:

a. $x^2 + y^2 = 5$

b. $x^2 + y^2 = 25$ ✓

c. $(x-5)^2 + y^2 = 0$

d. $y = \pm\sqrt{25 - x^2}$

Applying Foundations principles, $x^2 + y^2 = r^2 = 25$. Therefore, the verified correct option is $x^2 + y^2 = 25$.

39 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

A water tank (cylinder, radius 1 m) drains. If height decreases at 0.5 m/min, water volume leaving per minute:

a. 0.5π ✓

b. π

c. 2π

d. 4π

$V = \pi r^2 h \Rightarrow dV/dt = \pi(1)^2(-0.5) = -0.5\pi$ m³/min.

40 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

In $\hat{y} = a + bx$, the slope b represents:

a. The predicted value at x=0

b. **Change in y per unit x** ✓

c. Correlation coefficient

d. Standard error

Slope = rise over run; change in ■ per unit change in x.

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41 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$ ✓

c. $e^{-1} - 1$

d. $2 - e^{-1}$

The given differential equation for the y-component of the velocity is $y'' - 2y' + y = 0$. The characteristic equation is $r^2 - 2r + 1 = 0$, which factors as $(r-1)^2 = 0$. Thus, we have a repeated root $r = 1$. The general solution for $y(t)$ is $y(t) = c_1 e^t + c_2 t e^t$. Using the initial conditions $y(0) = 2$ and $y'(0) = 1$: $y(0) = c_1 e^0 + c_2(0)e^0 = c_1 = 2$. $y'(t) = 2e^t + 2te^t + c_2 e^t + c_2 t e^t = (2 + c_2)e^t + (2 + c_2)te^t$. $y'(0) = (2 + c_2)e^0 + (2 + c_2)(0)e^0 = 2 + c_2 = 1$, so $c_2 = -1$. Thus.

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42 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A team of statisticians is designing a complex experiment to evaluate the efficacy of a new fertilizer on crop yield across multiple regions and soil types. They have identified 3 distinct geographic regions and 5 different soil types. Within each region-soil type combination, they plan to test 4 different concentrations of the fertilizer. To account for variability, they will run 2 independent replicates of each treatment combination. However, due to resource limitations, they can only afford to run 150 unique treatment combinations in total. The team decides to allocate the available resources by prioritizing the combinations that are most likely to reveal significant differences, while ensuring a balanced representation across the major factors. They will select a subset of the possible treatment combinations. If they decide to use a stratified sampling approach where each region is a stratum, and within each region, they stratify by soil type, and then sample fertilizer concentrations, what is the minimum number of unique treatment combinations (region, soil type, concentration) they *must* include in their sample to ensure that every soil type is represented in at least one region's stratum, and every fertilizer concentration is represented within each soil type stratum across all regions, given that they still wish to have 2 replicates per selected combination?

a. 150

b. 90

c. 75

d. **60** ✓

The experiment has 3 regions, 5 soil types, and 4 fertilizer concentrations. This gives a total of $3 \times 5 \times 4 = 60$ unique treatment combinations (region, soil type, concentration). If they run 2 replicates for each of these combinations, they would need $60 \times 2 = 120$ experimental runs. The problem states they can afford 150 unique treatment combinations in total. However, the question asks for the minimum number of *unique treatment combinations* they must include to ensure representation. To ensur.

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43 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = \frac{a_n^2 + k}{2}$ for $n \geq 1$, where k is a positive constant. If the sequence converges to a limit L, prove that $L^2 - 2L + k = 0$. Furthermore, if the limit L is an integer and $k = 3$, determine the exact value of the limit.

a. **L = 3 ✓**

b. L = 1

c. L = -1

d. $L = \sqrt{3}$

If the sequence converges to a limit L, then as $n \to \infty$, $a_{n+1} \to L$ and $a_n \to L$. Substituting this into the recurrence relation, we get: $L = \frac{L^2 + k}{2}$ Multiplying by 2: $2L = L^2 + k$ Rearranging the terms to form a quadratic equation: $L^2 - 2L + k = 0$ This proves the first part of the problem. For the second part, we are given that $k = 3$ and the limit L is an integer. Substituting $k = 3$ into the equation derived above: $L^2 - 2L + 3 = 0$ We can solve this qua.

44 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider the vector field $\mathbf{F}(x, y) = \langle 2x, y \rangle$. Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ where C is the circle $x^2 + y^2 = 4$ oriented counterclockwise. Use Green's Theorem.

a. 0

b. 8π ✓

c. 4π

d. 16π

According to Green's Theorem, $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, where $\mathbf{F} = \langle P, Q \rangle$ and D is the region enclosed by C. ; Here, $P(x, y) = 2x$ and $Q(x, y) = y$. ; We compute the partial derivatives: $\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y) = 0$ and $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(2x) = 0$. ; Therefore.

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45 [Challenging] • Target: PCA & Dimensionality Reduction Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the matrix $A = (k, 2; 1, -3)$. If the matrix A is invertible and its inverse A^{-1} has a determinant of $-\frac{1}{11}$, find the value of k.

a. **4** ✓

b. 5

c. 3

d. 6

The determinant of matrix A is given by $\det(A) = (k)(-3) - (2)(1) = -3k - 2$. For an invertible matrix, its determinant is non-zero. The determinant of the inverse of a matrix is the reciprocal of the determinant of the original matrix, i.e., $\det(A^{-1}) = \frac{1}{\det(A)}$. We are given that $\det(A^{-1}) = -\frac{1}{11}$. Therefore, $-\frac{1}{11} = \frac{1}{-3k-2}$. This implies $-11 = -3k - 2$. Rearranging the equation to solve for k: $-11 + 2 = -3k$ $-9 = -3k$ $k = \frac{-9}{-3} = 3$. However, let's.

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